

HARRELL FIELD, AR, USA

WMO: 723425

Lat: 33.617N

Lon: 92.767W

Elev: 40

StdP: 100.84

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 00373

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.5	-4.1	-14.1	1.1	-1.6	-11.4	1.4	0.8	8.0	15.7	7.3	13.7	0.8	10	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	37.3	24.5	35.9	24.4	34.7	24.4	26.4	33.4	26.0	32.7	25.6	32.0	2.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	20.1	28.1	24.1	19.0	27.9	23.8	18.8	27.8	82.3	32.8	80.2	33.0	78.5	32.7	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.1	5.8	5.2	-10.3	38.8	3.0	1.2	-12.5	39.6	-14.2	40.3	-15.9	41.0	-18.1	41.9		
			-10.9	27.3	3.0	0.4	-13.1	27.6	-14.8	27.8	-16.5	28.0	-18.7	28.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	5.9	8.5	13.1	17.6	22.1	26.6	27.5	27.4	24.5	17.5	11.5	8.1
	DBStd	8.83	5.69	5.81	5.33	3.99	3.50	2.11	2.34	2.52	3.42	4.76	5.03	5.19
	HDD10.0	421	153	90	29	2	0	0	0	0	0	3	41	103
	HDD18.3	1518	387	277	176	61	11	0	0	0	1	74	211	321
	CDD10.0	3187	26	49	125	229	376	498	542	539	436	236	87	43
	CDD18.3	1242	2	3	14	38	128	248	284	281	187	49	6	2
	CDH23.3	13016	3	19	138	382	1201	2692	3101	3061	1832	526	52	9
	CDH26.7	6367	0	2	23	74	465	1393	1640	1646	919	196	9	0
Wind	WSAvg	1.8	2.0	2.2	2.4	2.2	1.9	1.7	1.5	1.4	1.3	1.4	1.7	1.9
Precipitation	PrecAvg	1379	116	108	128	127	138	97	97	71	96	134	102	131
	PrecMax	2111	235	254	303	424	327	206	264	253	266	547	254	224
	PrecMin	823	4	13	47	33	42	39	13	10	30	2	10	16
	PrecStd	318	56	61	66	93	78	50	71	51	63	115	63	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.0	26.1	28.8	30.0	33.8	37.5	37.8	38.9	37.5	33.9	27.9	24.8
		MCWB	15.1	18.0	18.7	19.4	22.6	23.7	25.0	25.4	23.6	21.9	19.7	19.0
	2%	DB	21.1	22.9	27.2	28.6	32.4	35.2	37.0	37.3	36.0	31.3	24.1	22.4
		MCWB	14.8	16.8	17.8	19.9	22.3	24.3	24.9	25.0	23.3	22.1	17.3	17.5
	5%	DB	18.0	21.2	24.8	27.4	31.1	34.0	35.1	35.2	33.9	28.8	22.5	19.9
		MCWB	13.7	16.0	16.6	19.1	22.2	24.0	24.9	24.8	23.3	21.3	17.0	16.3
	10%	DB	16.2	18.8	22.5	25.9	29.1	32.9	33.9	34.0	32.4	26.3	20.1	17.7
		MCWB	11.5	14.5	15.7	18.1	21.6	24.1	24.7	24.7	23.0	19.4	15.9	14.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.7	19.9	20.4	22.9	25.2	26.5	26.8	27.2	26.1	24.8	21.5	20.2
		MCDB	21.9	23.3	25.7	26.9	29.9	32.7	33.3	33.7	31.9	30.0	25.8	23.3
	2%	WB	17.0	18.7	19.2	21.6	24.1	25.9	26.3	26.5	25.3	23.7	19.9	18.9
		MCDB	19.3	22.1	24.3	26.2	29.2	32.2	33.4	33.7	31.3	28.8	22.7	21.4
	5%	WB	15.0	17.1	18.2	20.7	23.3	25.4	25.8	26.1	24.7	22.6	18.4	17.3
		MCDB	17.2	20.0	23.0	25.2	28.9	31.6	32.6	33.1	30.5	26.8	21.0	19.1
	10%	WB	12.1	15.2	16.9	19.6	22.5	24.9	25.5	25.6	24.0	21.1	16.6	14.9
		MCDB	15.5	18.4	20.9	24.0	27.9	30.9	32.0	32.1	29.6	24.5	19.6	17.2
Mean Daily Temperature Range	5% DB	MDBR	12.4	12.3	13.6	13.4	12.5	12.5	11.8	12.4	13.0	14.1	13.7	11.8
		MCDBR	16.1	14.8	15.1	14.6	14.3	13.7	13.4	14.2	15.7	15.8	15.4	14.8
		MCWBR	10.5	9.4	7.6	7.0	5.6	4.6	4.1	4.4	5.3	7.2	9.1	10.2
	5% WB	MCDBR	11.6	12.1	12.6	11.9	11.6	12.0	11.8	12.4	12.0	12.8	12.3	12.1
MCWBR		9.5	8.8	7.5	7.0	5.4	4.6	4.3	4.4	4.7	6.8	8.6	10.1	
Clear-Sky Solar Irradiance	taub	0.328	0.337	0.365	0.397	0.439	0.456	0.487	0.480	0.418	0.365	0.344	0.323	
	taud	2.506	2.480	2.419	2.328	2.268	2.258	2.199	2.208	2.359	2.480	2.491	2.529	
	Ebn at Noon	885	914	910	891	853	835	806	806	846	871	858	870	
	Edh at Noon	85	96	111	127	136	137	145	142	117	95	86	79	
All-Sky Solar Radiation	RadAvg	2.58	2.96	4.21	5.24	5.77	6.38	6.22	5.77	4.95	3.95	2.82	2.27	
	RadStd	0.25	0.48	0.36	0.34	0.46	0.52	0.39	0.48	0.43	0.54	0.33	0.26	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page